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#include<stdio.h>
#include<math.h>
#define maxsize 8
#define TRUE 1
#define FALSE 0

struct CQueue
{
int rear, front;
struct student{
int regn;
char name[20];
int marks;
}st[maxsize];
};

struct CQueue cq;

void insert();
void delete();

int empty();

void main()
{
char ch;
int c,x,i;
cq.rear=maxsize-1;
cq.front=maxsize-1;
do
{
printf("MENU\n");
printf("1. INSERT \n 2. DELETE\n 3. Display\n ");
scanf("%d",&c);
switch(c)
{
case 1: insert();

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        break;
case 2: delete();
        break;
case 3: printf("elements are\n");
        if(cq.front==maxsize-1)
        {
            for(i=0;i<=cq.rear;i++)
            {
                printf("%d\n",cq.st[i].regno);
                printf("%s\n",cq.st[i].name);
                printf("%d\n",cq.st[i].marks);
            }
        }
        else
        {
            for(i=cq.front+1;i<=cq.rear;i++)
            {
                printf("%d\n",cq.st[i].regno);
                printf("%s\n",cq.st[i].name);
                printf("%d\n",cq.st[i].marks);
            }
        }

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        break;
    }
    printf("do you wish to continue\n");
    scanf(" %c",&ch);
    }while(ch!='n');
}

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void insert()
{
    if(cq.front==((cq.rear+1)%maxsize))
    {
        printf("Queue overflow\n");
    }
}

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}  
else  
{cq.rear=(cq.rear+1)%maxsize;  
printf("enter the regno\n");  
scanf("%d",&cq.st[cq.rear].regno);  
printf("enter the name\n");  
scanf("%s",cq.st[cq.rear].name);  
printf("enter the marks\n");  
scanf("%d",&cq.st[cq.rear].marks);  
}  
}
```

```
int delete()  
{  
if(empty())  
{  
return(-1);  
}  
else  
{  
cq.front=(cq.front+1)%maxsize;  
printf("%d\n",cq.st[cq.front].regno);  
printf("%s\n",cq.st[cq.front].name);  
printf("%d\n",cq.st[cq.front].marks);  
}  
}  
int empty()  
{  
if (cq.rear==cq.front)  
return TRUE;  
else  
return FALSE;  
}
```